

Name: _____

RATIO AND PROPORTION, (SCALE FACTORS)

Date: _____

LO: I can use scale factors to find missing lengths in similar shapes and on maps.

Scale factors

A **scale factor** is the number you multiply by to enlarge or reduce a shape. It is the **ratio of corresponding lengths** between two similar shapes.

Finding it: Scale factor = new length $\div$ original length. If SF > 1 , the shape is enlarged. If SF < 1 , it is reduced.

Quick examples

●	SF = new $\div$ original = $12 \div 4 = 3$	enlarged by 3
●	Missing length = original $\times$ SF	multiply to find new
●	Area SF = (length SF) ²	area scales by square
●	SF 2 means add 2 to each side	multiply, not add
●	SF for area is same as for length	area uses SF ²

Key idea

Scale factor = new $\div$ original. Multiply any original length by the SF to get the new length.

$$\text{SF} = \frac{\text{new}}{\text{original}}$$

For area: $\times \text{SF}^2$. For volume: $\times \text{SF}^3$.

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – finding scale factor

Triangle A has sides 3, 4, 5 cm.
Triangle B has sides 9, 12, 15 cm.
Find the scale factor.

$$\text{SF} = 9 \div 3 = 3$$

$$\text{Check: } 12 \div 4 = 3 ; 15 \div 5 = 3$$

$$\text{Scale factor} = 3$$

Scale factor = 3

Divide any corresponding pair of sides.

Example 2 – finding missing length

Two similar rectangles. First: 5 cm by 8 cm. Second: 15 cm by ? cm.

$$\text{SF} = 15 \div 5 = 3$$

$$\begin{aligned} \text{Missing length} &= 8 \times 3 \\ &= 24 \text{ cm} \end{aligned}$$

24 cm

Find the SF from known corresponding sides, then multiply.

Example 3 – map scale

A map uses scale 1:50,000. A park is 4 cm on the map. Find real length.

$$1 \text{ cm} = 50,000 \text{ cm} = 500 \text{ m}$$

$$4 \text{ cm} = 4 \times 500$$

$$= 2000 \text{ m} = 2 \text{ km}$$

2 km

Multiply the map measurement by the scale factor.

Example 4 – area and scale factor

Two similar shapes have SF = 2. Small shape area = 15 cm². Find larger area.

$$\text{Area SF} = 2^2 = 4$$

$$\begin{aligned} \text{Larger area} &= 15 \times 4 \\ &= 60 \text{ cm}^2 \end{aligned}$$

60 cm²

Area scales by the square of the linear scale factor.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding scale factors

Find the scale factor from shape A to shape B.

- a) A: 4 cm, B: 12 cm
- b) A: 10 cm, B: 25 cm
- c) A: 6 cm, B: 2 cm
- d) A: 8 cm, B: 20 cm

2. Finding missing lengths

Use the scale factor to find the missing length.

- a) SF = 4. Original = 3 cm. New?
- b) SF = 2. Original = 7 cm. New?
- c) SF = $\frac{1}{2}$. Original = 16 cm. New?
- d) SF = 3. New = 21 cm. Original?

3. Similar shapes

Find the missing side in each pair of similar shapes.

- a) Triangles: 5, 12, 13 and 10, 24, ?
- b) Rectangles: 4 by 7 and 12 by ?
- c) Triangles: 6, 8, 10 and ?, 4, 5
- d) Rectangles: 9 by 15 and 6 by ?

4. Maps and models

Use the given scale to find real or map distances.

- a) 1:25,000. Map = 6 cm. Real?
- b) 1:50,000. Real = 3 km. Map?
- c) Model 1:100. Real building = 45 m. Model?
- d) 1:200,000. Map = 4.5 cm. Real?

5. Area and volume scale factors

Find areas or volumes using scale factors.

- a) Length SF = 3. Area of small = 8 cm². Large area?
- b) Length SF = 2. Volume of small = 10 cm³. Large volume?
- c) Areas: 9 cm² and 81 cm². Length SF?
- d) Volumes: 27 cm³ and 216 cm³. Length SF?
- e) Length SF = 4. Large area = 320 cm². Small area?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Scale factor = new length ÷ original length ☐

Correct answer: _____

Reason: _____

b) A scale factor of 2 means add 2 to each length ☐

Correct answer: _____

Reason: _____

c) Area scale factor = (length scale factor)² ☐

Correct answer: _____

Reason: _____

d) If the scale factor is $\frac{1}{3}$, the shape gets bigger ☐

Correct answer: _____

Reason: _____

e) Volume scale factor = (length scale factor)³ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Two similar cylinders have heights in ratio 2:5. The smaller has volume 80 cm³. Find the volume of the larger cylinder.

Expression: _____

Simplified: _____

b) On a map of scale 1:40,000, a lake has area 3.2 cm². Find the real area in m² and km².

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

